

Extraspecial-Group Sigma-MGFs

Odd exponent sectors and the D_8/Q_8 split

Computational verification note

17 July 2026

Abstract

We verify the proposed spectral formulas on all matrices of the two extraspecial groups of order 3^3 and on the faithful representations of D_8 and Q_8 . These examples are minimal but discriminating: they detect both the exponent- p versus exponent- p^2 split and the plus/minus Arf type. All 20 coefficient comparisons pass.

1 Odd p : what the character does not see

Let E be extraspecial of order p^{1+2n} and let ρ be a faithful irreducible representation of dimension $N = p^n$, with $\rho(z) = \omega I_N$. Central matrices contribute

$$p^{-2n} \Pi_p \prod_j (1 - t_j)^{-N}.$$

In the exponent- p case every noncentral element has all p th roots once per block of size p , hence

$$\mathcal{S}_{E, \exp p} = p^{-2n} \Pi_p \prod_j (1 - t_j)^{-N} + (1 - p^{-2n}) \prod_j (1 - t_j^p)^{-N/p}. \quad (1)$$

In the exponent- p^2 case, $g^p = z^{\ell(gZ)}$ for a nonzero linear functional $\ell : E/Z(E) \rightarrow \mathbb{F}_p$. Set $A_a(\mathbf{t}) = \prod_j (1 - \omega^a t_j^p)^{-N/p}$. Counting fibers of ℓ gives

$$\mathcal{S}_{E, \exp p^2} = p^{-2n} \Pi_p \prod_j (1 - t_j)^{-N} + (p^{-1} - p^{-2n}) A_0 + p^{-1} \sum_{a=1}^{p-1} A_a. \quad (2)$$

The coefficients in (2) sum to one. The last sum is the spectral information missed by the ordinary character, which vanishes off the center.

For the direct exponent-9 test we use $E = C_9 \rtimes C_3 = \langle x, y : y^{-1}xy = x^4 \rangle$ and the matrices

$$\rho(x) = \text{diag}(\eta, \eta^4, \eta^7), \quad \rho(y)e_j = e_{j-1}, \quad \eta = e^{2\pi i/9}.$$

The exponent-3 comparison is the 3-dimensional Heisenberg representation.

2 Extraspecial 2-groups

For sign $\varepsilon = +1$ (plus type) or -1 (minus type), put $N = 2^n$ and

$$A_+ = \prod_j (1 - t_j^2)^{-N/2}, \quad A_- = \prod_j (1 + t_j^2)^{-N/2}.$$

The quadratic form on $E/Z(E)$ has $2^{2n-1} + \varepsilon 2^{n-1}$ zeros and $2^{2n-1} - \varepsilon 2^{n-1}$ ones. Removing the identity coset from the first sector, restoring its two central lifts, and dividing by $|E|$ proves

$$\begin{aligned} \mathcal{S}_{E^\varepsilon} &= 2^{-2n} \Pi_2 \prod_j (1 - t_j)^{-N} \\ &\quad + \left(\frac{1}{2} + \varepsilon 2^{-n-1} - 2^{-2n}\right) A_+ + \left(\frac{1}{2} - \varepsilon 2^{-n-1}\right) A_-. \end{aligned} \quad (3)$$

For $n = 1$ the plus and minus representations are D_8 and Q_8 .

3 Direct coefficient test

The verifier constructs every representation matrix, computes $h_d(\text{spec } g)$ from $\det(I - tg)^{-1}$, and averages $\prod_i h_{\tau_i}$ over the group. It separately expands (1), (2), or (3). Selected outcomes are:

group	τ	direct	formula
exponent 3	(3)	2	2
exponent 9	(3)	1	1
exponent 3	(3, 3)	12	12
exponent 9	(3, 3)	11	11
D_8	(2)	1	1
Q_8	(2)	0	0
D_8	(1, 1)	1	1
Q_8	(1, 1)	1	1
D_8	(2, 2)	3	3
Q_8	(2, 2)	3	3

The unequal rows demonstrate that the tests would fail if the spectral sectors were collapsed. All 20 built-in cases pass to 2×10^{-8} .

4 Reproduction

```
.venv/bin/python for_this_guy/quantum_matrix_groups/
  extraspecial_groups/verification.py
.venv/bin/marimo edit for_this_guy/quantum_matrix_groups/
  extraspecial_groups/notebook.py
```

Join each displayed command onto one shell line.